Do insider purchases predict returns in real time? An interim report from a pre-registered forward test

stocktrack research · forward-test ledger started October 19, 2026

ABSTRACT

Twenty years of SEC Form 4 filings suggest that stocks bought by their own insiders beat comparable stocks by about +5.3% a year, but have not reliably beaten the S&P 500 since 2016 (+0.8% a year, not significant). Because those findings were shaped by looking at the same data, we test them forward: from October 19, 2026, a daily job has logged every qualifying open-market insider purchase before the market opens, then tracked its return against the S&P 500. By November 4, 2026 it had logged 161 company-days across 110 companies. The 91 buys with a completed first week beat the S&P 500 by +0.7% on average (median +1.7%; 64% ahead). Resampling whole entry days, the 95% interval is [−1.0%, +2.2%], which includes zero and contains the backtest's one-week average of +0.4%. No buy has yet reached the one-month or three-month horizons the hypotheses are judged on, so this report describes data collection and early noise, not evidence. Buys scoring 60 or more have not led so far (−1.4% against +2.0% for scores below 40, one week).

Keywords: insider trading, Form 4, forward test, pre-registration, event study

1. Introduction

Corporate insiders must report their trades in their own company's stock to the SEC within two business days, on Form 4. Purchases on the open market are the interesting ones: insiders sell for many reasons, but they buy mainly when they expect the price to rise. A long literature finds that insider purchases are followed by abnormal returns [1–3], and that the information is concentrated in non-routine, “opportunistic” trades [4].

This project began with a 20-year backtest of 395,205 purchase filings from 2006 to November 3, 2026. Its current run (November 4, 2026) finds that, over the three months after a filing, the stock beat an index of comparable surviving stocks by +5.3% a year (t 5.4), but beat the S&P 500 by only +0.8% a year since 2016 (p 0.83). Several kinds of buy looked stronger than average, such as purchases by chief financial officers, but they were found by looking at that data, so the data cannot also confirm them. Section 5 lists the hypotheses that were written down for a fresh test.

This paper reports that fresh test as it stands: what has been collected, how the first outcomes compare with the backtest, and when the registered hypotheses can be judged. It is an interim report; nothing in it is investment advice.

2. Design

Data. Every weekday morning (Tuesday to Saturday, 06:53 New York time) the job reads the previous business day's Form 4 filings from EDGAR and keeps open-market purchases (transaction code P) of exchange-listed shares priced at $2 or more that trade at least $1 million a day, excluding dividend-reinvestment and plan purchases.

Timing rule. A buy counts only if it was written to the ledger before 13:30 UTC on its entry day, the first session after the filing (the 09:30 open in summer, an hour before it in winter); the ledger records the timestamp. Buys logged later, for example ones filed before Monday's open, are kept but excluded (Section 4.5).

Outcome. The excess return is the stock's return from the entry-day open to the close 5, 21, 63, 126 and 252 sessions later, minus the S&P 500 ETF's (SPY) return over the same window. No trading costs are deducted and nothing is winsorized. Insiders buying the same company on the same day share one entry, so each company-day counts once, as in the backtest.

Groups. Each buy is tagged with the pre-registered backtest strategies it matches and with a live score (0–100). From October 24 each buy's signal-desk profile (above typical, typical or below typical) is also recorded at logging, so later refits cannot change what is judged.

Inference. Buys entered on the same day share that week's market moves, so they are not independent. Confidence intervals therefore resample whole entry days (10,000 draws); a naive interval that treats every buy as independent is shown for comparison and is too narrow.

3. The sample so far

Between October 19, 2026 and November 4, 2026 the ledger received 408 rows. Of these, 186 were logged in time to count, forming 161 company-days in 110 companies, bought by 142 different insiders, with entries from Oct 20 to Nov 3. A further 29 rows await their first session. Figure 1 shows the daily flow, about 15 company-days per session.

Directors make up the largest group of buyers, followed by 10% owners and chief executives (Table 1). 8% of the buys scored 60 or more on the live score, against 11% in the backtest's universe.

Table 1. Who bought, and how the live score graded the buys (company-days that count).

Buyer's role	Buys	Share	Live score	Buys	Share
CEO	18	11%	0-39	100	62%
CFO	6	4%	40-59	48	30%
Other exec	8	5%	60+	13	8%
President/Chair	4	2%			
Director	79	49%			
10% owner	41	25%			
Other	5	3%			

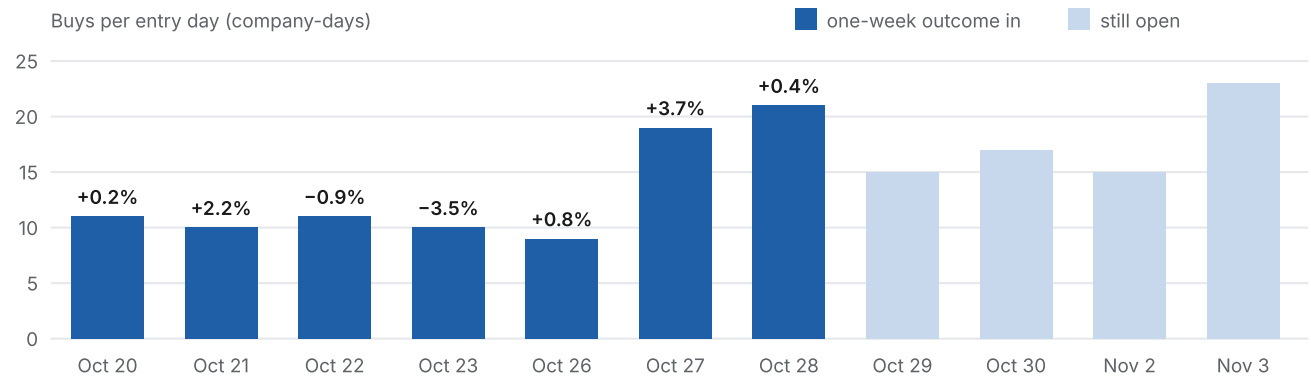


Figure 1. Company-days entered per session. The darker part has completed its first week; the label is that entry day's average one-week excess return over SPY. Buys entered on the same day share that week's market moves, which is why Table 2's interval resamples whole days.

4. Results

4.1 The first week

91 company-days, entered on 7 different days, have completed their first five sessions. Table 2 compares them with the backtest's one-week result for the same universe of liquid, discretionary buys. The live average is above the backtest's, and more buys finished ahead than in the backtest (64% against 51%). But the interval that respects the shared market moves is wide: it includes zero and the backtest value. With 7 entry days, the data cannot yet tell a true one-week edge of zero from one of a few percent.

Table 2. One-week excess return over SPY: live forward test against the 20-year backtest.

Sample	Buys	Mean	Median	Ahead	95% CI (by day)	95% CI (naive)
All buys, one week	91	+0.7%	+1.7%	64%	[−1.0%, +2.2%]	[−0.9%, +2.3%]
Backtest, same universe, one week	67,379	+0.4%	+0.1%	51%	–	–

Live returns are not winsorized; the backtest's are winsorized at 1% and 99%. The raw stock return over the same weeks averaged −0.1%, so SPY returned about −0.8% in those windows. Day-resampled interval from 7 entry days.

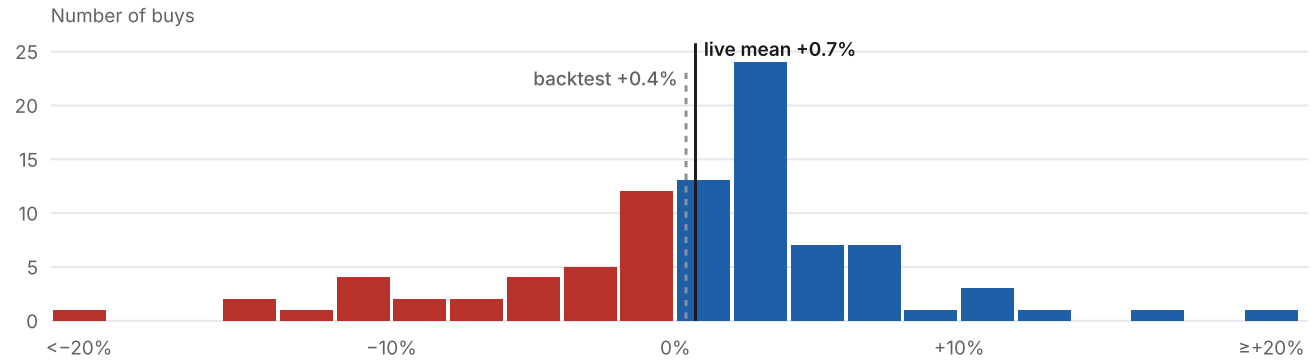


Figure 2. Distribution of one-week excess returns over SPY for every matured company-day, in 2-point bins; the outer bars gather everything beyond ±20%. The solid line is the live mean, the dashed line the backtest's.

4.2 The pre-registered strategies

Table 3 breaks the sample down by the strategies registered before the first backtest run. Most groups have only a handful of matured buys, so their one-week averages are dominated by single stocks. Figure 3 sets each group's live week against its backtest week. Among groups with at least five matured buys, the highest live averages so far belong to “Earlier buys beat SPY” (+3.1%) and “Opportunistic insiders” (+2.6%), and the lowest to “CEO or CFO, $100K+” (−5.4%) and “Stake up 25%+, $100K+” (−5.2%). One week of a three-month hypothesis is not evidence either way.

Table 3. Live results by pre-registered strategy. “Open” is the average excess return of all its buys from entry to the latest close.

Strategy	Logged	1-week n	1-week mean	Ahead	Open now	Backtest 1 week	Backtest 63 sessions
Every discretionary insider buy	161	91	+0.7%	64%	+1.5%	+0.4%	+0.8%
Live score 40+ (notable)	62	33	−1.6%	48%	−0.3%	+0.5%	+1.1%
Live score 60+ (strong)	15	9	−0.5%	56%	−0.7%	+0.6%	+1.6%

Strategy	Logged	1-week n	1-week mean	Ahead	Open now	Backtest 1 week	Backtest 63 sessions
CEO or CFO buying $100K+	14	7	−5.4%	43%	−1.5%	+0.6%	+1.4%
3+ insiders buying within 14 days	13	7	+0.6%	57%	+0.8%	+0.5%	+2.1%
Opportunistic insiders (Cohen-Malloy-Pomorski)	18	11	+2.6%	91%	+2.4%	+0.2%	+0.3%
Insider buying after a 30%+ drop	14	8	−2.8%	38%	+3.8%	+1.4%	+5.2%
Insiders whose earlier buys beat SPY	47	29	+3.1%	86%	+3.1%	+0.3%	+0.7%
Exec/director growing stake 25%+ with $100K+	17	7	−5.2%	14%	−0.1%	+0.4%	+0.9%
Exec's first buy in a year, $50K+	14	8	−0.1%	50%	+2.2%	+0.7%	+1.4%
CFO or other non-CEO executive buying $25K+ *(post-hoc)*	7	5	−0.6%	60%	+1.3%	+0.6%	+2.0%
Insider buying after a 30%+ run-up *(post-hoc)*	5	2	–	–	+2.6%	+0.8%	+2.6%

Means are shown from 3 matured buys. Post-hoc strategies were added after the first backtest run and are judged only on live data.

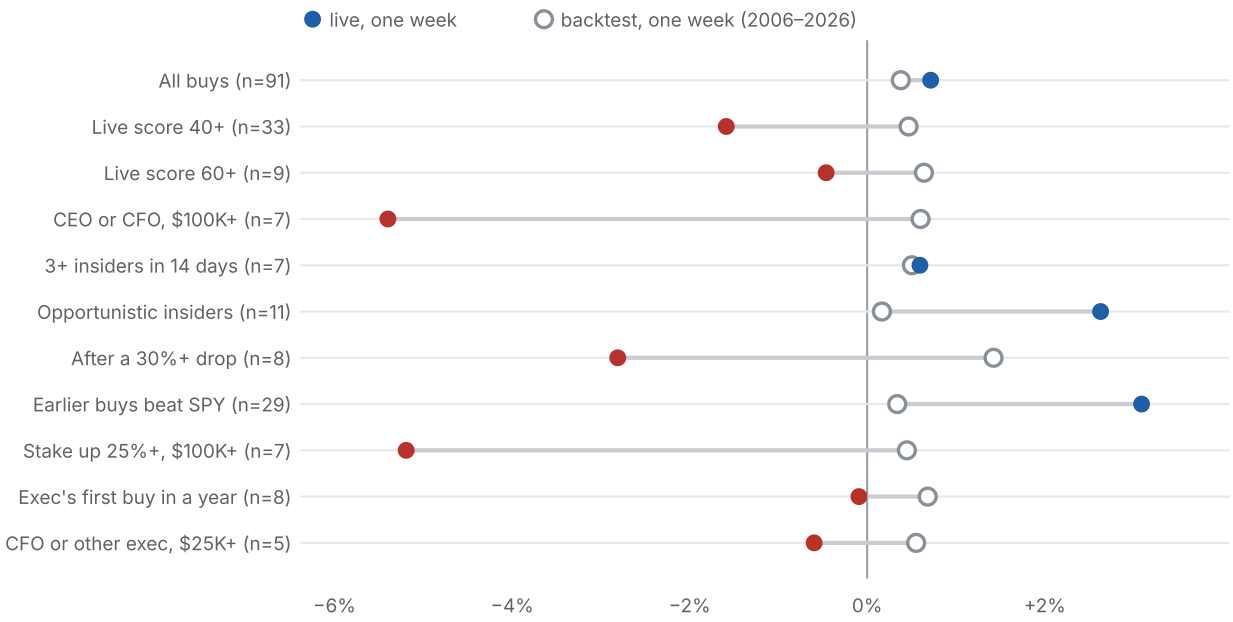


Figure 3. Live one-week mean (filled) against the backtest's one-week mean (ring) for groups with at least three matured buys.

4.3 Live score and signal-desk profile

If the score or the profile carried information, higher bands would lead. Buys scoring 60 or more have not led so far (−1.4% against +2.0% for scores below 40, one week). The bands hold few matured buys, and the profiles have only been recorded since October 24 (Table 4).

Table 4. One-week and open excess returns by live score band and by the signal-desk profile recorded at logging.

Band	Logged	1-week n	1-week mean	Ahead	Open now
Score 0-39	100	58	+2.0%	72%	+2.6%
Score 40-59	48	25	−1.6%	48%	0.0%
Score 60+	13	8	−1.4%	50%	−1.7%
Above typical	18	8	+3.8%	88%	+2.6%
Typical	65	25	+2.0%	84%	+0.4%
Below typical	35	15	+0.8%	67%	+2.3%

Profiles count each company-day once, with its largest buy, as the registered tier hypothesis does.

4.4 Open positions

All 161 company-days that have entered are marked to their latest close. After an average of 5 sessions they are +1.5% ahead of SPY on average (median +1.5%; 66% ahead). These overlapping, unequal holding periods are not a test of anything, but they show where the first one-month outcomes, due from November 17, 2026, are heading.

4.5 Buys logged too late to count

163 company-days were logged after their entry session had opened, most of them on the first day, when the job back-filled filings from late September and early October. They are excluded from every result above. For completeness: 162 have one-week outcomes averaging −1.4%, and 78 have one-month outcomes averaging +1.0% (62% ahead).

5. The registered hypotheses

Four hypotheses are on record in data/backtest/hypotheses.json. Each is judged once 100 of its buys have completed 63 sessions: it is supported if the mean excess return over SPY is positive and its 95% interval excludes zero. No buy has completed 63 sessions yet. Table 5 projects, from the rate at which each group has been logging buys, when it will reach 100 buys and when the last of them will mature.

Table 5. Registered hypotheses, their expected effect, and when they can be judged at the current logging rate.

Hypothesis	Expected	Since	Logged	Per session	100 logged by	Judged after
CFO or other non-CEO executive buying $25K+ (post-hoc)	+1% to +2% excess over 63 sessions vs SPY; less after small-cap underperformance	Oct 21	10	1.00	March 17, 2027	June 16, 2027
Insider buying after a 30%+ run-up (post-hoc)	+2% to +4% excess over 63 sessions, with wide dispersion	Oct 21	5	0.50	August 10, 2027	November 8, 2027
Owners filing as "Other" and routine insiders	below the all-buys forward average	Oct 21	20	2.00	January 4, 2027	April 6, 2027
Above-typical vs below-typical profiles	a small gap or none; recent years showed none	Oct 24	21	2.62	December 18, 2026	March 23, 2027

Projections assume the logging rate stays as it has been; they will move. The tier hypothesis is gated by its smaller group.

6. Discussion

Too early to judge. One week is the shortest horizon the ledger tracks, and the hypotheses are about three months. Over a single week, company news and market swings dwarf any plausible edge: the backtest's one-week average is +0.4%, while live buys range from −40.8% to +26.5%.

One market, one month. Every outcome so far comes from a few weeks in October and November 2026. Small companies, which most insider buys are, can lead or lag the S&P 500 for months at a time, and that alone can push every group the same way. The entry-day interval in Table 2 is the honest measure of how little is known.

Costs. Returns are before trading costs. Under the backtest's cost model a round trip costs 0.2% to 1.5% for the middle 80% of these buys, depending on price and trading volume, which matters next to one-week edges of a percent or two.

Data quirks. Some filings coded as open-market purchases are not ordinary trades. Several large buys were at prices far from the market's (marked ‡ in the appendix), which suggests an offering, a private placement or another share class; some buyers are institutions adding to closed-end funds; and some officer titles read only “See Remarks”. These rows follow the pre-registered rules and stay in the sample.

7. Conclusion

The forward test is working as designed: it logs 15 qualifying company-days a session, before the open, and tracks them against SPY. The first week's results are consistent with the backtest and with no edge at all. The first one-month outcomes arrive on November 17, 2026 and the first three-month ones on January 20, 2027. At the current pace, the last of the registered hypotheses can be judged around November 8, 2027.

References

1. Seyhun, H. N. (1986). Insiders' profits, costs of trading, and market efficiency. *Journal of Financial Economics*, 16(2), 189–212.
2. Lakonishok, J., & Lee, I. (2001). Are insider trades informative? *Review of Financial Studies*, 14(1), 79–111.
3. Jeng, L. A., Metrick, A., & Zeckhauser, R. (2003). Estimating the returns to insider trading: A performance-evaluation perspective. *Review of Economics and Statistics*, 85(2), 453–471.
4. Cohen, L., Malloy, C., & Pomorski, L. (2012). Decoding inside information. *Journal of Finance*, 67(3), 1009–1043.

Appendix: the largest one-week moves

Table 6. The best and worst one-week excess returns among matured company-days.

Ticker	Company	Buyer	Bought	Entry	1 week vs SPY
KRRO	Korro Bio, Inc.	10% Owner	$885K	Oct 27	+26.5%
XFOR	X4 Pharmaceuticals, Inc	Executive Chairman	$12K	Oct 20	+16.6%
OIG	Orion180 Insurance Group Inc.	10% Owner	$116K	Oct 27	+13.1% ‡
HQY	HEALTHEQUITY, INC.	EVP Chief Prod & Strat Off	$12K	Oct 20	+10.9%
GPI	GROUP 1 AUTOMOTIVE INC	10% Owner	$24M	Oct 27	+10.9%
LXU	LSB INDUSTRIES, INC.	Director	$10K	Oct 27	+10.7%
AIOK	AI OKTO Corp.	Chief Executive Officer	$1.1M	Oct 28	−40.8% ‡
RAPP	Rapport Therapeutics, Inc.	Chief Financial Officer	$144K	Oct 22	−14.8%
AGMB	AgomAb Therapeutics NV	Director	$89K	Oct 23	−14.4%
RAPP	Rapport Therapeutics, Inc.	Director	$249K	Oct 23	−13.3%
CNXU	Conexeu Sciences Inc.	Director	$500K	Oct 20	−11.6%
IAM	Iambic Therapeutics, Inc.	Director	$12M	Oct 20	−11.6%

‡ The insider's price was more than 15% from the prior close: likely an offering, private placement or another share class.

REPRODUCING THIS PAPER

Every number comes from the repository's committed files: `data/picks.csv` (the ledger), `data/backtest/latest.json` and `data/backtest/hypotheses.json`. Regenerate with `python3 -m stocktrack research paper` (this copy: commit e67defe). This is research, not investment advice.